

Transcriptional Analysis of IncF_{repB}-Mediated *bla*_{OXA-48}-Positive Plasmid Characterized from *Escherichia coli* ST448

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Objectives: To investigate the transcriptional response of *bla*_{OXA-48} and the copy number alteration of IncF_{repB} plasmid carrying *bla*_{OXA-48} under an antibiotic concentration gradient.

Methods: *Escherichia coli* strains harboring *bla*_{OXA-48} on an IncF_{repB} plasmid were isolated from Silchar Medical College and Hospital, Silchar, India. Sequence type and common resistance determinants were determined by PCR assay. Plasmid copy number alteration and the transcriptional expression of *bla*_{OXA-48} under different antibiotic pressures were determined by quantitative real-time PCR, and the relative fold change was measured by the $\Delta\Delta CT$ method.

Results and Conclusion: The plasmid that carried *bla*_{OXA-48} in *E. coli* ST448 was characterized as IncF_{repB} and found to be conjugatively transferable. The isolates were found to coexist with *bla*_{NDM-1} within the IncX3-type plasmid. It was observed that the copy number and transcriptional response of *bla*_{OXA-48} were directly proportional to the increasing concentration of meropenem and ertapenem, whereas in the case of imipenem, it was reversed. The identification of *bla*_{OXA-48} through IncF_{repB}-type plasmid in this study indicates the potential route of spread of this resistance determinant in this area and also the insights we gained from the transcriptional changes of *bla*_{OXA-48} in response to different antibiotic pressures could also facilitate the development of novel or alternative therapeutic options needed for multidrug-resistant infections.

Keywords: *bla*_{OXA-48}, *Escherichia coli* ST448, copy number, transcriptional response

Background

AFTER THE FIRST report from Turkey in 2001, *bla*_{OXA-48} has been increasing in a number of reports of OXA-48-producing organisms throughout the world, and the plasmid-mediated occurrence of *bla*_{OXA-48} has led to the dissemination of this resistance determinant not only in hospitals but also in different ecosystems, imposing a serious problem.¹ In the earlier studies, it has been reported that OXA-48-producing isolates are increasingly identified from different enterobacterial isolates in many parts of the world such as North Africa, the Middle East, and the European Countries, and in the majority of the cases, dissemination of this gene was found to be associated with a self-transferable IncL/M plasmid.^{2–4} OXA-48 and its variants are unique carbapenemases with low-level hydrolytic activity toward carbapenems and also show very weak activity against expanded-spectrum cephalosporins; however, isolates are frequently detected as multidrug resistant harboring multiple resistance determinants.⁵ In the present study, we have investigated the expression of *bla*_{OXA-48} and the alteration in the copy number

of the plasmid under gradient antibiotic pressure concentration. A part of this work has been presented in the joint meeting of ISAAR and ICIC 2017 (11th International Symposium on Antimicrobial Agents and Resistance and 3rd International Interscience Conference on Infection and Chemotherapy 2017) held in Busan, Korea.

Methods

Bacterial strains

Two multidrug-resistant *Escherichia coli* isolates harboring *bla*_{OXA-48}, that is, DP-Ec101 and Dp-Ec209, were identified in a retrospective study of carbapenem-resistant *E. coli* isolates from Silchar Medical College and Hospital, Silchar, India, which is the only tertiary referral center in the southern part of Assam. DP-Ec101 was collected from the urine culture of a 69-year-old lady suffering from urinary tract infection and was admitted to the medicine ward, and the second strain Dp-Ec209 was isolated from a clinical specimen of endotracheal aspirate of a hospitalized

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female patient of 26 years admitted to the intensive care unit of the same hospital.

Coexistence of other resistance determinants

Co-occurrence of class A (*bla*_{KPC}, *bla*_{IM/NMC}, and *bla*_{SME}), class B (*bla*_{NDM}, *bla*_{VIM}, and *bla*_{IMP}), and other class D (*bla*_{OXA-23} and *bla*_{OXA-58}) carbapenemases was determined by the performance of different PCR assays as described earlier.^{6,7} In addition, the coexistence of extended-spectrum β -lactamases (ESBLs) targeting *bla*_{TEM}, *bla*_{SHV}, *bla*_{CTX-M}, *bla*_{OXA-2}, *bla*_{OXA-10}, *bla*_{PER}, *bla*_{VEB}, and *bla*_{GES} was also determined.⁸ The amplified product of the resistance determinant was purified using the MinElute PCR Purification Kit (Qiagen, Germany) and then sequenced to confirm the variant of the resistant determinants.

Sequence typing

The sequence type of the isolates was determined by multilocus sequence typing, targeting the internal fragments of seven conserved housekeeping genes of *E. coli*. Later, the STs were assigned based on the existing *E. coli* database.⁷ The details of the housekeeping genes and the oligonucleotides used in this study are mentioned in Table 1.

Susceptibility testing

Minimum inhibitory concentrations of the isolates harboring *bla*_{OXA-48} were determined by the agar dilution method to determine the susceptibility profile of these strains against imipenem (Merck, France), meropenem (AstraZeneca, United Kingdom), ertapenem (MSD France), amikacin (Zuche pharmaceuticals, India), cefepime, aztreonam (Aristo, India), piperacillin/tazobactam (Alkem, India), and polymyxin-B (Samarth, India). Each stock solution for the corresponding antibiotic was made at 1 mg/mL concentration in nuclease-free water and stored at -80°C . The quality control for the stock solution was checked each time against *E. coli* ATCC 25922. The result of the susceptibility testing was interpreted as per the Clinical and Laboratory Standards Institute (CLSI) guidelines.⁹

Conjugation assay

The experiment was carried out using clinical strains of *E. coli* coharboring *bla*_{OXA-48} and *bla*_{NDM-1} as a donor and azide-resistant *E. coli* J53 as the recipient. Both the donor and recipient cells were cultured in Luria Bertani broth (Hi-Media) until it reaches an optical density of 0.8–0.9 at 600 nm (OD₆₀₀). Cells were mixed at a ratio of 1:5 donor-to-recipient, and the inoculated plates were incubated overnight at 37°C . Transconjugants were selected on Luria Bertani medium containing imipenem (1 mg/L) and sodium azide (1,000 mg/L) as described earlier. Plasmids were extracted from the selected transconjugants using the QIAprep Spin Miniprep Kit (Qiagen), and PCR assay was performed using the primer sequences of *bla*_{OXA-48} and *bla*_{NDM-1} (Table 1) to identify the specific resistance determinant carried within the transconjugants.

Plasmid incompatibility typing

The incompatibility type of the conjugatively transferable plasmid carrying *bla*_{OXA-48} or *bla*_{NDM-1} was determined by

PCR-based replicon typing targeting 18 different replicon types, namely, FIA, FIB, FIC, HI1, HI2, I1/I γ , L/M, N, P, W, T, A/C, K, B/O, X, Y, F, and FIIA. In addition, different groups of IncX-types such as IncX1, IncX2, IncX3, and IncX4 were also targeted.¹⁰ The plasmid encoding the transconjugant was extracted using the QIAprep Spin Miniprep Kit (Qiagen) and was used as a template for the reaction, and the oligonucleotides that were used to identify these replicon types are mentioned in Table 1.

Plasmid stability analysis

Stability of the plasmid harboring *bla*_{OXA-48} in wild strains and transconjugants was performed by the serial passage method consecutively for 100 days (wild strains) and 40 days (transconjugants), respectively, at 1:1,000 dilutions without any antibiotic pressure. The bacterial suspension was inoculated into 10 mL LB broth and incubated at 37°C in a shaker incubator at 200 rpm for 12 hours and the steps repeated with a freshly prepared broth at 1:1,000 dilutions until the harboring carbapenemase gene got lost. After each passage, 1 mL of the culture was diluted in physiologic saline (1:1,000), and 40 μL of the diluted sample was spread on to the LB agar plate. Following overnight incubation, 50 colonies from the agar plates were randomly picked and subjected to PCR assay to confirm the presence of *bla*_{OXA-48}. The loss of the resistance determinants was confirmed by performing the experiment in triplicate.¹¹

Plasmid copy number determination

Copy number alteration of transconjugant plasmid carrying *bla*_{OXA-48} was determined by real-time quantitative PCR using the Step One Plus Real-Time detection system (Applied Biosystem, Warrington, UK).¹¹ A single colony was inoculated into the LB broth containing 0.5, 1, 2, and 4 $\mu\text{g/mL}$ of imipenem, meropenem, ertapenem, and oxacillin, and also without any antibiotic (considered a reference for the reaction), and incubated at 37°C for 4–7 hours until the OD reaches 0.9 at A₆₀₀. The plasmid was extracted from each condition using a QIAprep Spin Miniprep Kit (Qiagen) and was used as a template. The reaction was carried out using 10 mL of SYBR[®] Green PCR Master Mix (Applied Biosystem), 4 ng plasmid DNA as a template, and 3 mL of each primer (10 picomol) in a 20 mL reaction under a reaction condition of initial denaturation at 94°C for 5 minutes followed by 40 cycles of denaturation at 94°C for 20 seconds, annealing at 52°C for 40 seconds, and extension at 72°C for 30 seconds. The relative fold change was measured by the $\Delta\Delta\text{CT}$ method, and the CT value of each sample was normalized against the housekeeping gene *rpsLE* of *E. coli*. Each set of reactions was run in triplicate, and the experiment was repeated thrice.

Transcriptional expression analysis of *bla*_{OXA-48}

Transcriptional expression of *bla*_{OXA-48} responds to four different concentrations of imipenem, meropenem, ertapenem, and oxacillin stress (0.5, 1, 2, and 4 $\mu\text{g/mL}$) and was determined by inoculating the organisms in Luria Bertani broth (Hi-media, Mumbai, India). After 12 hours of incubation, total RNA was

TABLE 1. OLIGONUCLEOTIDES USED IN THIS STUDY

Target	Gene	Sequence	Product size	Reference
OXA-48	Class D carbapenemase	OXA-48-F: 5'-GATTATCGGAATGCCTGCGG-3'	845	8
OXA-23		OXA-48-R: 5'-CTACAAGCGCATCGAGCATCA-3'		
		OXA-23-F: 5'-GATCGGATTGGAGAACCAGA-3'	501	
OXA-58		OXA-23-R: 5'-ATTCTGACCGCATTTCAT-3'		
		OXA-58-F: 5'-CGATCAGAATGTTCAAGCGC-3'	529	
		OXA-58-R: 5'-ACGATTCTCCCTCTGCGC-3'		
NDM	Class B carbapenemase	NDM-F: 5'-GGGCAGTCGCTTCCAACGGT-3'	476	6
VIM		NDM-R: 5'-GTAGTGCTCAGTGTGCGCAT-3'		
		VIM-F: 5'-GATGGTGTGTTGGTTCGCATA-3'	390	
IMP		VIM-R: 5'-CGAATGCGCAGCACCAG-3'		
		IMP-F: 5'-TTGACACTCCATTACDG-3'	139	
		IMP-R: 5'-GATYGAGAATTAAGCCACYCT-3'		
KPC	Class A carbapenemase	KPC-F: 5'-CATTCAAGGGCTTTCTTGCTGC-3'	538	6
IMI/NMC		KPC-R: 5'-ACGACGGCATAGTCATTTGC-3'		
		IMI/NMC-F: 5'-CCATTCAACCCATCACAAC-3'	440	
SME		IMI/NMC-R: 5'-CTACCGCATAATCATTGTC-3'		
		SME-F: 5'-AACGGCTTCATTTTTGTTTAG-3'	831	
		SME-R: 5'-GCTTCCGCAATAGTTTTATCA-3'		
CTX-M	Extended-spectrum β -lactamase genes	CTX-M-F: 5'-CGCTTTGCGATGTGCAG-3'	550	7
SHV		CTX-M-R: 5'-ACCGCGATATCGTTGGT-3'		
		SHV-F: 5'-AGGATTGACTGCTTTT-3'	392	
OXA-2		SHV-R: 5'-ATTTGCTGATTTTCGCTCG-3'		
		OXA-2-F: 5'-AAGAAACGCTACTCGCCTGC-3'	478	
		OXA-2-R: 5'-CCACTCAACCCATCCTACCC-3'		
OXA-10		OXA-10-F: 5'-TCAACAAATCGCCAGAGAAG-3'	276	
		OXA-10-R: 5'-TCCCACACCAGAAAAACCAG-3'		
TEM		TEM-F: 5'-ATGAGTATTCAACATTTCCG-3'	867	
PER		TEM-R: 5'-CTGACAGTTACCAATGCTTA-3'		
		PER-F: 5'-AATTTGGGCTTAGGGCAGAA-3'	920	
		PER-R: 5'-ATGAATGTCATTATAAAAGC-3'		
VEB		VEB-F: 5'-CATTTCCCGATGCAAAGCGT-3'	650	
		VEB-R: 5'-CGAAGTTTCTTTGACTCTG-3'		
GES		GES-F: 5'-AGTCGGCTAGACCGGAAAG-3'	863	
		GES-R: 5'-TTTGTCCGTGCTCAGGAT-3'		
adk		adk F: 5'-ATTCTGCTTGGCGCTCCGGG-3'	583	8
fumC		adk R: 5'-CCGTCAACTTTCGCGTATTT-3'		
		fumC F: 5'-TCACAGGTCCGAGCGCTTC-3'	806	
icd	Escherichia coli housekeeping genes	fumC R: 5'-GTACGCAGCGAAAAAGATTTC-3'		
		icd-F: 5'-ATGGAAAGTAAAGTAGTTGTTCCGGCACA-3'	878	
		icd-R: 5'-GGACGCGACGAGGATCTGTT-3'		
purA		purA-F: 5'-CGCGCTGATGAAAGAGATGA-3'	816	
		purA-R: 5'-CATACGGTAAGCCACGCAGA-3'		
gyrB		gyrB-F: 5'-TCGGCGACACGGATGACGGC-3'	911	
		gyrB-R: 5'-ATCAGGCCTTCACGCGCATC-3'		
recA		recA-F: 5'-CGCATTGCTTTACCTGACC-3'	780	
		recA-R: 5'-TCGTCGAAATCTACGGACCGGA-3'		
mdh		mdh-F: 5'-ATGAAAGTCGCAGTCCTCGGCGCTGCTGGCGG-3'	932	
		mdh-R: 5'-TTAACGAACTCCTGCCCCAGAGCGATATCTTTCTT-3'		
HI1		HI1-F: 5'-GGAGCGATGGATTACTTCAGTAC-3'	471	10
HI2	Replicon types	HI1-R: 5'-TGCCGTTTCACTTCGTGAGTA-3'		
		HI2-F: 5'-TTTCTCCTGAGTCACCTGTTAACAC-3'	644	
I1		HI2-R: 5'-GGCTCACTACCGTTGTCATCCT-3'		
		I1-F: 5'-CGAAAGCCGGACGGCAGAA-3'	139	
		I1-R: 5'-TCGTCTGTTCCGCCAAGTTCGT-3'		
X		X-F: 5'-AACCTTAGAGGCTATTTAAGTTGCTGAT-3'	376	
		X-R: 5'-TGAGAGTCAATTTTATCTCATGTTTAGC-3'		
L/M		L/M-F: 5'-GGATGAAAATATCAGCATCTGAAG-3'	785	
		L/M-R: 5'-CTGCAGGGGCGATTCTTTAGG-3'		
N		N-F: 5'-GTCTAACGAGCTTACCGAAG-3'	559	
		N-R: 5'-GTTTCAACTCTGCCAAGTTC-3'		
FIA		FIA-F: 5'-CCATGCTGGTTCTAGAGAAGGTG-3'	462	
		FIA-R: 5'-GTATATCCTTACTGGCTTCCGCAG-3'		
FIB		FIB-F: 5'-GGAGTTCTGACACACGATTTTCTG-3'	702	
		FIB-R: 5'-CTCCCGTCGCTTCAGGGCATT-3'		
W		W-F: 5'-CCTAAGAACAACAAAGCCCCCG-3'	242	
		W-R: 5'-GGTGCGCGCATAGAACCGT-3'		
Y		Y-F: 5'-AATTCAAACAACACTGTGCAGCCTG-3'	765	
		Y-R: 5'-GCGAGAATGGACGATTACAAAACCTT-3'		
P		P-F: 5'-CTATGGCCCTGCAAACGCGCCAGAAA-3'	534	
		P-R: 5'-TCACGCGCCAGGGCGCAGCC-3'		

(continued)

TABLE 1. (CONTINUED)

Target	Gene	Sequence	Product size	Reference
FIC		FIC-F: 5'-GTGAAGTGGCAGATGAGGAAGG-3' FIC-R: 5'-TTCTCCTCGTCGCCAAACTAGAT-3'	262	10
A/C		A/C-F: 5'-GAGAACCAAGACAAAGACCTGGA-3' A/C-R: 5'-ACGACAAACCTGAATTGCCTCCTT-3'	465	
T		T-F: 5'-TTGGCCTGTTTGTGCCTAAACCAT-3' T-R: 5'-CGTTGATTACACTTAGCTTTGGAC-3'	750	
FIIS		FIIS-F: 5'-CTGTCGTAAGCTGATGGC-3' FIIS-R: 5'-CTCTGCCACAAACTTCAGC-3'	270	
F _{repB}		F _{repB} -F: 5'-TGATCGTTTAAGGAATTTTG-3' F _{repB} -R: 5'-GAAGATCAGTCACACCATCC-3'	270	
K/B		K/B-F: 5'-GCGGTCCGGAAAGCCAGAAAAC-3'	160	
K		K-R: 5'-TCTTTCACGAGCCCGCCAAA-3'		
B/O RV		B/O-R: 5'-TCTGCGTTCGCCAAGTTCGA-3'	159	
X1	Inc X replicon types	X1-F: 5'-GCTTAGACTTTGTTTTATCGTT-3' X1-R: 5'-TAATGATCCTCAGCATGTGAT-3'	461	10
X2		X2-F: 5'-GCGAAGAAATCAAAGAAGCTA-3' X2-R: 5'-TGTTGAATGCCGTTCTTGTCAG-3'	678	
X3		X3-F: 5'-GTTTCTCCAGCCTTGTTCA-3' X3-R: 5'-CTTTGTGCTTGCTATCATAA-3'	351	
X4		X4-F: 5'-AGCAAACAGGGAAAGGAGAAGACT-3' X4-R: 5'-TACCCCAAATCGTAACCTG-3'	569	

isolated using the Qiagen RNeasy Mini Kit (Qiagen), immediately reverse transcribed into cDNA by using the QuantiTect[®] Reverse Transcription Kit (Qiagen), and the cDNA was quantified by Picodrop (Pico 200; Cambridge, United Kingdom). Quantitative real-time PCR was performed using the Power Sybr Green Master Mix (Applied Biosystem) in the Step One Plus Real-Time detection system (Applied Biosystem). The housekeeping gene *rpsLE* of *E. coli* was used as an internal standard. The relative expression of *bla*_{OXA-48} and the antibiotic pressure were determined by the $\Delta\Delta C_t$ method.¹²

Results

Both the *E. coli* strains carrying *bla*_{OXA-48} belonged to the same sequence type ST448, and these strains were also found to coharbor *bla*_{NDM-1}. Susceptibility testing revealed that these isolates showed resistance toward amikacin, cefepime, piperacillin/tazobactam, imipenem, meropenem, and ertapenem, leading to a multidrug-resistant phenotype, but were found susceptible only to tigecycline and polymyxin (Table 2). Like parent strains, transconjugants carrying *bla*_{OXA-48} were also found to have a similar pattern of susceptibility profile showing resistance to all the tested antibiotics

except polymyxin and tigecycline (Table 2). The *bla*_{OXA-48} resistance determinant was carried through a conjugatively transferable IncF_{repB}-type plasmid of ~70 kb in size, but interestingly, *bla*_{NDM-1} was found to be located in a different 90 kb plasmid of IncX3 type. Stability analysis of the parental strains revealed that the resistant plasmids were stable until the 90th serial passage, but interestingly, the *bla*_{OXA-48} gene was found lost after the 65th serial passage, whereas *bla*_{NDM-1} carrying plasmid was found stable until the 90th serial passage. Significantly, in the case of transconjugants carrying *bla*_{OXA-48}, the stability was drastically reduced, and the plasmid was maintained only for 30th serial passage, while the transconjugant carrying *bla*_{NDM-1} was stable until the 40th passage (Supplementary Fig. S1).

Real-time PCR analysis revealed that the copy number and transcriptional expression of *bla*_{OXA-48} were directly proportional to the concentration of meropenem and ertapenem, that is, with the higher concentration of these antibiotics, the relative copy number and the transcriptional response of *bla*_{OXA-48} also increased, whereas in the case of imipenem, it was inversely proportional and was found to be reduced with the increasing concentration of imipenem. However, under the oxacillin pressure, the expression

TABLE 2. MINIMUM INHIBITORY CONCENTRATIONS OF CLINICAL ISOLATES OF *ESCHERICHIA COLI* AND TRANSCONJUGANTS CARRYING *BLA*_{OXA-48}

Strain ID Antibiotics	DP-Ec101		Dp-Ec209		E. coli J53
	Parent strain (mg/L)	Transconjugants (mg/L)	Parent strain (mg/L)	Transconjugants (mg/L)	
Imipenem	64	16	64	16	0.06
Meropenem	64	32	64	32	0.06
Ertapenem	128	32	32	32	0.06
Amikacin	64	16	128	16	0.125
Cefepime	128	64	128	64	0.125
Piperacillin/tazobactam	64	64	64	64	0.125
Tigecycline	0.5	0.125	0.5	0.125	0.006
Polymyxin-B	1	0.125	1	0.5	0.006

and copy number of OXA-48 were found to be variable with the concentration of antibiotic (Fig. 1). Transcriptional analysis revealed that the expression of $\text{bla}_{\text{OXA-48}}$ was induced under the increasing concentration of meropenem, ertapenem, and oxacillin, whereas in the case of imipenem pressure, the expression level was found to be downregulated. The expressional pattern of $\text{bla}_{\text{OXA-48}}$ was found specific for each concentration of the antibiotics.

Discussion

The production of carbapenemases severely compromises treatment options, and the increased spread of these resistance determinants is considered a major public health threat.¹³ Earlier, Turkey, the Middle East, and the North African countries were considered a reservoir of $\text{bla}_{\text{OXA-48}}$.¹⁴ However, the present study reporting this resistant determinant's occurrence within nosocomial isolates of *E. coli* from an Indian hospital underscores its wide dissemination in this subcontinent. In India, $\text{bla}_{\text{OXA-48}}$ was earlier detected in *Klebsiella pneumoniae* and *Enterobacter cloacae*,^{15,16} and this indicates that the gene is being maintained in a diverse range of hosts. The coexistence of multiple resistance genes within an isolate is exceptionally problematic as it minimizes therapeutic alternatives, and this may depend on their transfer by mobile genetic elements and stable plasmids with and without selective pressure in clinical settings. The high-stability nature

of the plasmid encoding $\text{bla}_{\text{OXA-48}}$ indicates the potential ability to be maintained within the host for several generations, even after the withdrawal of antibiotic pressure.¹⁷ In earlier studies, it was recorded that the $\text{bla}_{\text{OXA-48}}$ gene was customarily linked to the IncL/M-type plasmid. In contrast, the present study reports the $\text{IncF}_{\text{repB}}$ type as a carrier of $\text{bla}_{\text{OXA-48}}$, which warrants further investigation to track the origin and acquisition. The antibiotic susceptibility profile of the parent strains and the transconjugants carrying $\text{bla}_{\text{OXA-48}}$ was found to be in a similar pattern in this study, which is in contrast to the previous reports where the weak activity of $\text{bla}_{\text{OXA-48}}$ toward carbapenems and cephalosporins was observed and indicates the growing resistance pattern over a period of time.⁵

This study also shows that antibiotic pressure increases the copy number of the plasmid. In the case of ertapenem pressure, the copy number alteration was found unique compared with other carbapenems. So, maintaining this high copy number of the plasmid within the cell under some selection pressure may create an impact on the fitness and survival of the harboring organism. However, no previous study is found to support our data.

The current study shows that the spread of the $\text{bla}_{\text{OXA-48}}$ gene is propagated through $\text{IncF}_{\text{repB}}$ -type plasmid in India, which underscores the necessity of systematic screening of hospitalized patients and early detection of these multidrug-resistant bacteria. Also, the insights that we have gained from the transcriptional changes of $\text{bla}_{\text{OXA-48}}$ in response to

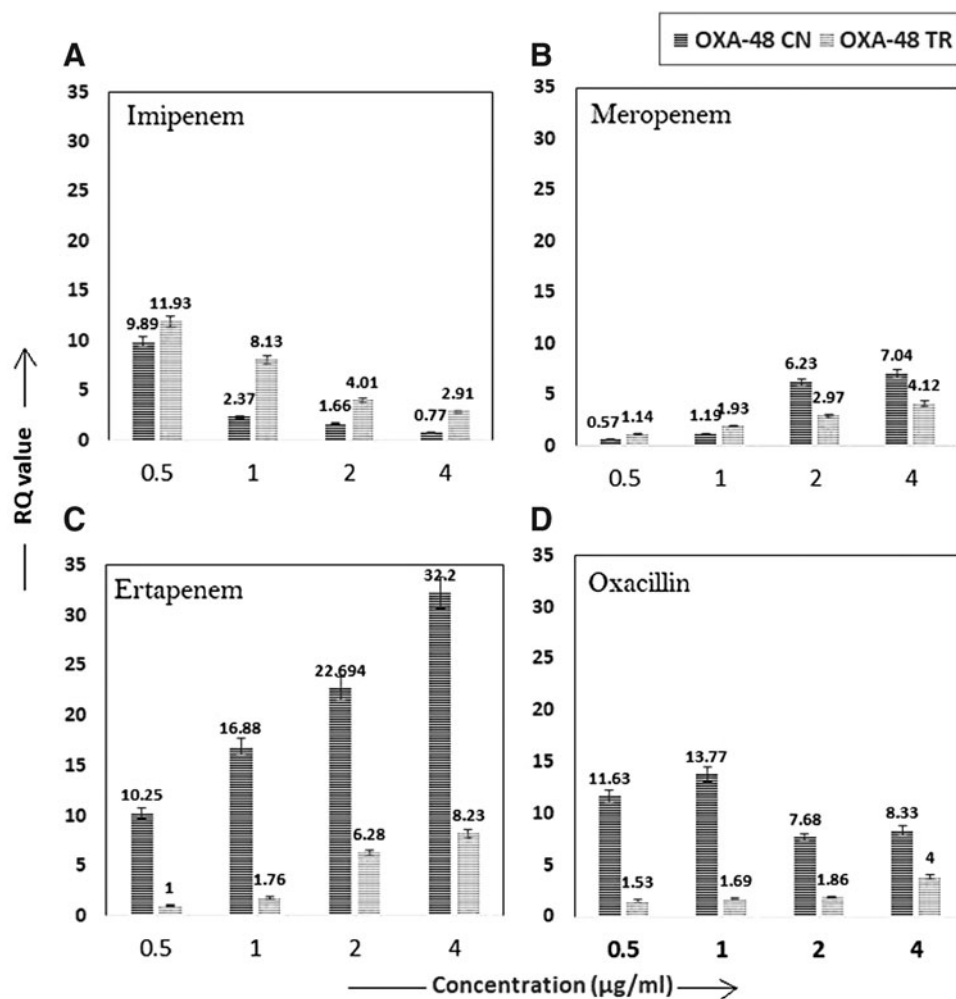


FIG. 1. (A) Imipenem, (B) Meropenem, (C) Ertapenem, (D) Oxacillin. Copy number (OXA-48 CN) and transcriptional response (OXA-48 TR) of $\text{bla}_{\text{OXA-48}}$ under different antibiotic pressures.

different antibiotic pressures can facilitate the development of novel or alternative therapeutic options that are needed for multidrug-resistant infections.

Ethical Approval

The work was approved by Institutional Ethics committee of Assam University, Silchar vide Reference Number: IEC/AUS/C/2014–001.

Availability of Data and Materials

All the relevant data and information are presented in the article.

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Authors' Contributions

D.P.: Performed the experimental work, data collection and analysis, and prepared the article. D.D. and A.C.: Participated in experiment designing and article correction. A.B. Supervised the research work and participated in designing the study and drafting the article. All authors read and approved the final article.

Disclosure Statement

No competing financial interests exist.

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Supplementary Material

Supplementary Figure S1

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